

Sequences & Series

ANSWER KEY



The binomial series for negative and fractional powers

- 1 Expand $(1+x)^{-1}$ up to the term in x^3 , using $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3$.
With $n = -1$: $1 + (-1)x + \frac{(-1)(-2)}{2}x^2 + \frac{(-1)(-2)(-3)}{6}x^3 = 1 - x + x^2 - x^3$.
- 2 Expand $(1+x)^{1/2}$ up to the term in x^3 .
With $n = \frac{1}{2}$: coefficient of x^2 is $\frac{\frac{1}{2}(-\frac{1}{2})}{2} = -\frac{1}{8}$; coefficient of x^3 is $\frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{6} = \frac{1}{16}$. So
 $(1+x)^{1/2} \approx 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$.
- 3 Expand $(1-x)^{-3}$ up to the term in x^2 .
Write $(1+u)^{-3}$ with $u = -x$: coefficient of u is -3 , coefficient of u^2 is $\frac{(-3)(-4)}{2} = 6$. So
 $1 - 3(-x) + 6(-x)^2 = 1 + 3x + 6x^2$.
- 4 Expand $(1+x)^{-4}$ up to the term in x^2 .
With $n = -4$: coefficient of x is -4 ; coefficient of x^2 is $\frac{(-4)(-5)}{2} = 10$. So $(1+x)^{-4} \approx 1 - 4x + 10x^2$.

Validity conditions for the binomial series

- 5 State the range of values of x for which the binomial expansion of $(1+x)^{-2}$ is valid.
The expansion of $(1+x)^n$ for non-integer or negative n is valid for $|x| < 1$, i.e. $-1 < x < 1$.
- 6 State the range of values of x for which the expansion of $(3+x)^{-1}$ converges, given that after factoring, the general form requires $\left|\frac{x}{3}\right| < 1$.
 $\left|\frac{x}{3}\right| < 1 \Rightarrow |x| < 3$.

Using series expansions to approximate values

- 7 Use $(1+x)^{1/2} \approx 1 + \frac{x}{2} - \frac{x^2}{8}$ to approximate $\sqrt{1.02}$, to four decimal places.
With $x = 0.02$: $1 + 0.01 - \frac{0.0004}{8} = 1 + 0.01 - 0.00005 = 1.00995 \approx 1.0100$.
- 8 Use $(1+x)^{-1} \approx 1 - x + x^2 - x^3$ to approximate $\frac{1}{1.05}$, to four decimal places.
With $x = 0.05$: $1 - 0.05 + 0.0025 - 0.000125 = 0.952375 \approx 0.9524$, matching the exact value $\frac{1}{1.05} \approx 0.9524$.
- 9 Use the geometric series $(1-x)^{-1} \approx 1 + x + x^2 + x^3$ (the binomial series with $n = -1$) to approximate $\frac{1}{0.98}$, to four decimal places.
Writing $\frac{1}{0.98} = (1-x)^{-1}$ with $x = 0.02$: $1 + 0.02 + 0.0004 + 0.000008 \approx 1.0204$, matching the exact value $\frac{1}{0.98} \approx 1.020408$.

- 10 Use $(1+x)^{-1} \approx 1 - x + x^2$ to approximate $\frac{1}{1.01}$, to four decimal places.
With $x = 0.01$: $1 - 0.01 + 0.0001 = 0.9901$.

Expansions of the form $(a+bx)^n$

- 11 Expand $(4+x)^{1/2}$ up to the term in x^2 , by writing $(4+x)^{1/2} = 4^{1/2} \left(1 + \frac{x}{4}\right)^{1/2}$.
 $4^{1/2} = 2$. $\left(1 + \frac{x}{4}\right)^{1/2} \approx 1 + \frac{x}{8} - \frac{x^2}{128}$. So $(4+x)^{1/2} \approx 2 \left(1 + \frac{x}{8} - \frac{x^2}{128}\right) = 2 + \frac{x}{4} - \frac{x^2}{64}$.
- 12 Expand $(2-3x)^{-1}$ up to the term in x^2 , by writing $(2-3x)^{-1} = 2^{-1} \left(1 - \frac{3x}{2}\right)^{-1}$, and state the range of validity.
 $\left(1 - \frac{3x}{2}\right)^{-1} \approx 1 + \frac{3x}{2} + \frac{9x^2}{4}$. So $(2-3x)^{-1} \approx \frac{1}{2} \left(1 + \frac{3x}{2} + \frac{9x^2}{4}\right) = \frac{1}{2} + \frac{3x}{4} + \frac{9x^2}{8}$. Validity requires $\left|\frac{3x}{2}\right| < 1$, i.e. $|x| < \frac{2}{3}$.
- 13 Expand $(1+2x)^{-2}$ up to the term in x^2 , and state the range of validity.
With $n = -2$, $u = 2x$: coefficient of u is -2 , coefficient of u^2 is $\frac{(-2)(-3)}{2!} = 3$. So
 $1 - 2(2x) + 3(2x)^2 = 1 - 4x + 12x^2$. Validity: $|2x| < 1$, i.e. $|x| < \frac{1}{2}$.
- 14 Expand $(9-x)^{1/2}$ up to the term in x^2 , by writing $(9-x)^{1/2} = 9^{1/2} \left(1 - \frac{x}{9}\right)^{1/2}$.
 $9^{1/2} = 3$. $\left(1 - \frac{x}{9}\right)^{1/2} \approx 1 - \frac{x}{18} - \frac{x^2}{648}$. So $(9-x)^{1/2} \approx 3 \left(1 - \frac{x}{18} - \frac{x^2}{648}\right) = 3 - \frac{x}{6} - \frac{x^2}{216}$.

Constructing Maclaurin series

- 15 Find the Maclaurin series for $f(x) = \cos x$ up to the term in x^4 , using
 $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$.
 $f(0) = 1$, $f'(0) = -\sin 0 = 0$, $f''(0) = -\cos 0 = -1$, $f'''(0) = \sin 0 = 0$, $f^{(4)}(0) = \cos 0 = 1$. Series:
 $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$.
- 16 Find the Maclaurin series for $f(x) = e^{2x}$ up to the term in x^3 .
 $f(0) = 1$. $f'(x) = 2e^{2x}$, $f'(0) = 2$. $f''(x) = 4e^{2x}$, $f''(0) = 4$. $f'''(x) = 8e^{2x}$, $f'''(0) = 8$. Series:
 $1 + 2x + \frac{2^2}{2}x^2 + \frac{2^3}{6}x^3 = 1 + 2x + 2x^2 + \frac{4}{3}x^3$.
- 17 Find the Maclaurin series for $f(x) = \ln(1-x)$ up to the term in x^3 .
 $f(0) = 0$. $f'(x) = -\frac{1}{1-x}$, $f'(0) = -1$. $f''(x) = -\frac{1}{(1-x)^2}$, $f''(0) = -1$. $f'''(x) = -\frac{2}{(1-x)^3}$, $f'''(0) = -2$.
Series: $-x - \frac{x^2}{2} - \frac{x^3}{3}$.
- 18 Find the Maclaurin series for $f(x) = \sin(2x)$ up to the term in x^3 , using $\sin u \approx u - \frac{u^3}{6}$ with $u = 2x$.
 $\sin(2x) \approx 2x - \frac{(2x)^3}{6} = 2x - \frac{8x^3}{6} = 2x - \frac{4}{3}x^3$.

Building new series from known ones

- 19 Using $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$, find the Maclaurin series for e^{-x^2} up to the term in x^4 , by substituting $-x^2$ for x .
 Substituting $u = -x^2$ into $1 + u + \frac{u^2}{2}$ (the cubic term in u would give x^6 , beyond the required order):
 $1 + (-x^2) + \frac{(-x^2)^2}{2} = 1 - x^2 + \frac{x^4}{2}$.
- 20 Given $\sin x \approx x - \frac{x^3}{6}$ and $(1 + u)^{-1} \approx 1 - u + u^2$, find the Maclaurin series for $\frac{1}{1 + \sin x}$ up to the term in x^2 (using $u = \sin x \approx x$ to this order).
 Treating $u = \sin x \approx x$ up to x^2 order: $\frac{1}{1 + \sin x} \approx 1 - x + x^2$.
- 21 Using $\frac{1}{1-x} \approx 1 + x + x^2 + x^3$, differentiate term by term to find the Maclaurin series for $\frac{1}{(1-x)^2}$ up to the term in x^2 .
 Differentiating the left side gives $\frac{1}{(1-x)^2}$; differentiating the right side term by term: $1 + 2x + 3x^2 + \dots$.
 So $\frac{1}{(1-x)^2} \approx 1 + 2x + 3x^2$.
- 22 Using $\ln(1+x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}$, find the Maclaurin series for $\ln(1+2x)$ up to the term in x^3 , by substituting $2x$ for x .
 Substituting $u = 2x$: $\ln(1+2x) \approx 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} = 2x - 2x^2 + \frac{8}{3}x^3$.

Using series to approximate definite integrals

- 23 Use $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$ to approximate $\int_0^{0.5} \cos x \, dx$, to four decimal places, and compare with the exact value $\sin(0.5) \approx 0.4794$.
 Integrating term by term:
 $\left[x - \frac{x^3}{6} + \frac{x^5}{120} \right]_0^{0.5} = 0.5 - \frac{0.125}{6} + \frac{0.03125}{120} \approx 0.5 - 0.020833 + 0.00026 \approx 0.4794$, matching $\sin(0.5)$ closely.
- 24 Given the standard result $(1-x^2)^{-1/2} \approx 1 + \frac{x^2}{2} + \frac{3x^4}{8}$, approximate $\int_0^{0.2} (1-x^2)^{-1/2} \, dx$, to four decimal places, and compare with the exact value $\arcsin(0.2) \approx 0.2014$.
 Integrating term by term:
 $\left[x + \frac{x^3}{6} + \frac{3x^5}{40} \right]_0^{0.2} = 0.2 + \frac{0.008}{6} + \frac{3(0.00032)}{40} \approx 0.2 + 0.001333 + 0.000024 \approx 0.2014$, matching $\arcsin(0.2)$ closely.

A well-designed word problem: relativistic momentum correction

- 25 This question has three parts. In special relativity, the momentum correction factor is $\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$, where v is speed and c is the speed of light. (a) Using the binomial expansion with $u = -\frac{v^2}{c^2}$, show that $\gamma \approx 1 + \frac{v^2}{2c^2} + \frac{3v^4}{8c^4}$ for small $\frac{v^2}{c^2}$. (b) A satellite travels at $v = 7000$ m/s, with $c = 3 \times 10^8$ m/s. Using only the first-order approximation $\gamma \approx 1 + \frac{v^2}{2c^2}$, find $\gamma - 1$, to three significant figures, in standard form. (c) Explain, by comparing orders of magnitude, why the second term $\frac{3v^4}{8c^4}$ is negligible for this satellite compared to the first-order term.

(a) With $u = -\frac{v^2}{c^2}$ and $|u| < 1$:
 $\left(1 + u\right)^{-1/2} \approx 1 - \frac{1}{2}u + \frac{3}{8}u^2 = 1 - \frac{1}{2}\left(-\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^2}{c^2}\right)^2 = 1 + \frac{v^2}{2c^2} + \frac{3v^4}{8c^4}$. (b)
 $\frac{v^2}{c^2} = \frac{(7000)^2}{(3 \times 10^8)^2} = \frac{4.9 \times 10^7}{9 \times 10^{16}} = 5.44 \times 10^{-10}$. (c) The second term equals $\frac{3}{8}\left(\frac{v^2}{c^2}\right)^2 \approx \frac{3}{8}(5.44 \times 10^{-10})^2 \approx 1.11 \times 10^{-19}$, roughly nine orders of magnitude smaller than the first-order term (2.72×10^{-10}) — since $\frac{v}{c}$ is already extremely small, squaring the ratio in the second term makes it astronomically smaller still, so it can safely be ignored.